25901130 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBOT200 - BIOFERTILIZERS AND BIOCONTROL AGENTS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **1** marks

1. List any two objectives of minimal cultivation. [K] / [CO1]
2. What are bio-fertilizers? [K] / [CO2]
3. Why is Rhizobium inoculation ineffective for cereal crops like wheat? [K] / [CO2]
4. Which specific gene in Bt is responsible for producing insecticidal protein crystals? [K] / [CO3]
5. How biopesticides protect plants ? [U] / [CO3]
6. What is microbial inoculation in composting? [K] / [CO5]
7. How active ingredients in bioformulations influence their activity? [K] / [CO4]
8. Name one benefit of using biofertilizers. [K] / [CO6]
9. Name a fungi that is used as a biocontrol agent. [K] / [CO4]

10. Name one enzyme involved in composting. [K] / [CO5]

11. Name any two nitrogen-fixing microorganisms used as bio-fertilizers. [K] / [CO2]

12. What is the function of fungi in compost? [K] / [CO5]

[1x10 = 10]

Part B

Short Essay

Answer any **6** questions

Each question carries **5** marks

13. Give a brief account on sustainable agriculture. [U] / [CO1]

14. Do we need to promote the production and use of biopesticides? Justify your answer. [An] / [CO3]

15. How does Trichoderma contribute to soil health and the decomposition of organic matter? [U] / [CO3]

16. Write an account on bioformulations used for the uptake of Phosphorus [K] / [CO4]

17. Write a comparative account on wettable powder and dust type of bioformulations [U] / [CO4]

18. Write briefly about the advantages of using Trichoderma instead of chemical pesticides. [U] / [CO6]

19. Give a brief account on potash-solubilising bio-fertilizers. [U] / [CO2]

20. Describe biodegradable waste and its importance in compost production. [U] / [CO5]

[5x6 = 30]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

21. Discuss ectomycorrhiza as a biofertilizer. [U] / [CO2]

22. Explain the isolation, mass multiplication, and formulation of Rhizobium as a biocontrol agent. [U] / [CO6]

[10x1 = 10]